

Migration

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Summary prepared for DFA Milestone 1 Exam Preparation

Core Research Question

How does the movement of stocks between portfolios contribute to the size premium (small stocks outperforming large) and value premium (value stocks outperforming growth)?

Methodology

Portfolio Construction

Six portfolios formed annually each June from 1926 to 2005, using independent sorts on size and book-to-market ratios:

- **Size groups:** Small (S) = below NYSE median market cap; Big (B) = above NYSE median
- **Book-to-market groups:** Growth (G) = top 30%; Neutral (N) = middle 40%; Value (V) = bottom 30%
- **Six portfolios created:** SG, SN, SV (small growth/neutral/value) and BG, BN, BV (big growth/neutral/value)

Migration Categories

Each year, stocks in a portfolio can experience four types of outcomes:

- **Same:** Stocks staying in identical portfolio year-over-year
- **Plus:** Stocks improving (moving toward growth or acquired by another firm)
- **Minus:** Stocks deteriorating (moving toward value, delisted for cause, or negative book equity)
- **dSize:** Stocks changing size category (small→big or big→small)

Key Findings: Size Premium

The Surprising Result: The size premium comes almost entirely from small stocks that become big stocks.

Migration Type	Frequency	Excess Return	Contribution
Small → Big	8-12% of market cap	+50% to +62%	+5 to +9 ppt/year
Big → Small	1-2% of market cap	-31% to -37%	Minimal (~-0.5 ppt)

What doesn't drive the size premium: Big→small transitions have strongly negative returns but affect only 1-2% of big stock market cap. The Same, Plus, and Minus groups actually lean against the size premium, contributing more to big stock returns than to small stock returns.

Key Findings: Value Premium

The value premium arises from three distinct sources:

1. Plus Transitions (Contribution: 3.2-3.7 ppt/year)

- Value stocks frequently improve toward neutral/growth (20-25% of market cap annually)
- Growth stocks rarely improve further (only 1-3% of market cap can move higher)
- Both groups earn high returns (~16-22%) when improving
- Value benefits more due to higher frequency of positive transitions

2. Minus Transitions (Contribution: 1.2-5.1 ppt/year)

- Growth stocks frequently deteriorate toward value (11-26% of market cap)
- Value stocks rarely deteriorate further (0.1-1% of market cap)
- Both groups earn terrible returns (-12% to -36%) when deteriorating
- Much larger impact for small stocks (5.1% contribution) vs. big stocks (1.2%)

3. Same Category (Contribution: 1.0-1.7 ppt/year)

Value stocks that stay value earn modestly higher returns than growth stocks that stay growth. This contributes through return differential rather than frequency differences.

What Reduces the Value Premium:

Small growth stocks move to big more frequently (11.8% of market cap) than small value (8.5%). Both earn spectacular returns (~61%) when becoming big. This greater frequency for growth stocks reduces the small stock value premium by 2.9% annually.

Value Premium Decomposition

Source	Small Stocks	Big Stocks
Plus transitions	+3.7 ppt	+3.2 ppt
Minus transitions	+5.1 ppt	+1.2 ppt
Same (non-migrants)	+1.0 ppt	+1.7 ppt
dSize (S→B frequency)	-2.9 ppt	N/A
Total Value Premium	~7.0 ppt	~5.7 ppt

Transition Patterns

Persistence

Most stocks remain in their original portfolio from year to year:

- Big growth: 87% of market cap stays (most stable portfolio)
- Big value and big neutral: ~75% of market cap stays
- Small portfolios: ~60% of market cap stays (least stable)
- Type changes (value↔growth) are more common than size changes (small↔big)

Delisting Patterns

- **Good delists (mergers):** 1-3% of stocks annually, earning +16% to +35% excess returns
- **Bad delists (failure):** Rare for big stocks (<0.2%), more common for small (1-3%), earning -18% to -54%
- **Surprising finding:** Small growth stocks have highest rates of negative book equity (1.7% vs 0.4% for value)

Economic Interpretation

The paper presents two competing explanations for the size and value premiums:

Rational Pricing View

Prices reflect correct probabilities of transitions and appropriate risk premiums. The size and value premiums compensate investors for bearing genuine economic risks. When portfolios are formed, market prices already incorporate the best possible forecasts of (i) transition probabilities and (ii) future prices resulting from those transitions.

Behavioral Mispricing View

Investors systematically underestimate the probabilities that: (i) small stocks become big, (ii) value stocks improve in type, and (iii) growth stocks deteriorate. This causes unexpected returns when these transitions occur more frequently than anticipated. The consistently irrational (and learning-impaired) investors who dominate prices fail to correctly predict migration patterns.

Note: The paper frames but does not resolve this debate. The migration patterns are consistent with both rational risk-based and behavioral mispricing explanations.

Bottom Line

Size Premium

The size premium is remarkably concentrated—it comes almost entirely from small winners that grow big. This high-return phenomenon affects only 8-12% of small stock market cap annually, yet these transitions contribute 5-9 percentage points to small stock returns. Without this single migration type, there would be essentially no size premium.

Value Premium

The value premium is more balanced and arises from three complementary sources:

- Value stocks' higher propensity to improve in type (Plus transitions)
- Growth stocks' higher propensity to deteriorate (Minus transitions)
- Modestly better returns for value stocks that don't migrate (Same category)

These positive contributions are partially offset by small growth stocks' greater tendency to become big (which carries very high returns but occurs more frequently for growth than value).

Key Implications for DFA Investment Philosophy

- **Patient capital advantage:** Value stocks require time to realize improvements in type—turnover costs would destroy the Plus transition premium
- **Small stock capacity constraints:** The size premium's concentration in small winners that grow big implies limited capacity in small cap strategies
- **Trading around transitions:** High returns are associated with transitions, but these are unpredictable ex-ante; broad exposure is essential
- **Empirical support for factor premiums:** Migration patterns provide mechanical explanation for why size and value premiums exist and persist

Study Period and Data

This analysis covers 80 years (1926-2005 portfolio formations, 1927-2006 returns) and includes NYSE, AMEX (post-1962), and NASDAQ (post-1972) stocks. The remarkable consistency of findings across this long period and through multiple market regimes strengthens confidence in the results.